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Differentiating x^n

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Thankfully, there's a quick way to differentiate terms of the form x^n (where n is a constant) with having to use first principles every time:

! If $y = ax^n$ then $\frac{dy}{dx} = nax^{n-1}$ (where a, n are constants)
i.e. multiply by the power and reduce the power by 1

Examples:

$$y = x^5$$

$$f(x) = x^{\frac{1}{2}}$$

$$y = 2x^6$$

$$f(x) = \frac{x}{x^4}$$

$$y = \sqrt{x^6}$$

Why would it be incorrect to say that $y=2^x$ differentiates to $\frac{dy}{dx}=x2^{x-1}$?
The rule only works when the base is x and the power is a constant. Neither is true here! Note that x^n is "a power of x" whereas 2^x is an exponential term (which you will encounter more in Chp14), and therefore differentiate differently. You will learn how to differentiate exponential terms in Year 2.

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Test Your Understanding

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1. $y = x^7$
2. $y = 3x^{10}$
3. $f(x) = \frac{x^{\frac{1}{2}}}{x^2}$
4. $y = ax^a$
5. $f(x) = \sqrt{49x^7}$

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Differentiating Multiple Terms

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Differentiate $y = x^2 + 4x + 3$

First thing to note:
If $y = f(x) + g(x)$ then
$\frac{dy}{dx} = f'(x) + g'(x)$
i.e. **differentiate each term individually in a sum/subtraction**.

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Quickfire Questions

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1. $y = 2x^2 - 3x$
2. $y = 4 - 9x^3$
3. $y = 5x + 1$
4. $y = ax$

(where a is a constant)

5. $y = 6x - 3 + px^2$

(where p is a constant)

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Harder Example

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Let $f(x) = 4x^2 - 8x + 3$
a) Find the gradient of $y = f(x)$ at the point $\left(\frac{1}{2}, 0\right)$
b) Find the coordinates of the point on the graph of $y = f(x)$ where the gradient is 8.
c) Find the gradient of $y = f(x)$ at the points where the curve meets the line $y = 4x - 5$.

a

b

c

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Test Your Understanding

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Let $f(x) = x^2 - 4x + 2$
a) Find the gradient of $y = f(x)$ at the point $(1, -1)$
b) Find the coordinates of the point on the graph of $y = f(x)$ where the gradient is 5.
c) Find the gradient of $y = f(x)$ at the points where the curve meets the line $y = 2 - x$.

a

b

c

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5 Find the gradients of the curve $y = x^2 - 5x + 10$ at the points A and B where the curve meets the line $y = 4$.

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6 Find the gradients of the curve $y = 2x^2$ at the points C and D where the curve meets the line $y = x + 3$.

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7\quad $\mathrm{f}(x) = x^2 - 2x - 8$.
a Sketch the graph of $y = \mathrm{f}(x)$.
b On the same set of axes, sketch the graph of $y = \mathrm{f}'(x)$.
c Explain why the x-coordinate of the turning point of $y = \mathrm{f}(x)$ is the same as the x-coordinate of the point where the graph of $y = \mathrm{f}'(x)$ crosses the x-axis.

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Differentiating Harder Expressions

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If your expression isn't a sum of x^n terms, simply manipulate it until it is!

1. Turn roots into powers: $y = \sqrt{x}$

$y = \frac{1}{\sqrt[3]{x}}$

2. Split up fractions. $y = \frac{x^2+3}{\sqrt{x}}$

3. Expand out brackets. $y = x^2\,(x - 3)$

4. Beware of numbers in denominators! $y = \frac{1}{3x}$

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Test Your Understanding

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Differentiate the following.

$$y = \frac{1}{\sqrt{x}}$$

$$y = \frac{2+x^3}{x^2}$$

$$y = \frac{1+2x}{3x\sqrt{x}}$$

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Exercise 12E – Pearson Pure Y1 page 267-268

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6\quad $\mathrm{f}(x) = \frac{12}{p\sqrt{x}} + x$, where p is a real constant and $x > 0$. Given that $\mathrm{f}'(2) = 3$, find p, giving your answer in the form $a\sqrt{2}$ where a is a rational number.

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Exercise 12E – Pearson Pure Y1 page 267-268

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7\quad $\mathrm{f}(x) = (2 - x)^9$.
a Find the first 3 terms, in ascending powers of x, of the binomial expansion of $\mathrm{f}(x)$, giving each term in its simplest form.
b If x is small, so that x^2 and higher powers can be ignored, show that $\mathrm{f}'(x) \approx 9216x - 2304$.

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Extension

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The expression $\frac{d^2}{dx^2}[(2x-1)^4(1-x)^5] + \frac{d}{dx}[(2x+1)^4(3x^2-2)^2]$

is a polynomial of degree:

9

8

7

less than 7

This just means “differentiate twice”. The ‘second derivative’ comes later in this chapter.

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Second Order Derivatives

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When you differentiate once, the expression you get is known as the **first derivative**. Unsurprisingly, when we differentiate a second time, the resulting expression is known as the **second derivative**. And so on…

Original Function	First Derivative	Second Derivative
$y = x^4$ (Leibniz's)	$\frac{dy}{dx} = 4x^3$	$\frac{d^2y}{dx^2} = 12x^2$
$y = x^4$	$y\prime = 4x^3$	$y\prime\prime = 12x^2$
$y = x^4$ (Newton's)	$\dot{y} = 4x^3$	$\ddot{y} = 12x^2$
$f(x) = x^4$ (Lagrange's)	$f\prime(x) = 4x^3$	$f\prime\prime(x) = 12x^2$

You can similarly have the third derivative ($\frac{d^3y}{dx^3}$), although this is no longer in the A Level syllabus. We'll see why might use the second derivative soon…

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Just for your interest…

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How does the notation $\frac{d^2y}{dx^2}$ work?
Why are the squareds where they are?

Suppose that $y = x^3 + 1$.
Then when we write $\frac{dy}{dx}$, we're effectively doing $\frac{d(x^3+1)}{dx}$ (by substitution), although this would typically be written:

$$\frac{d}{dx}\left(x^3 + 1\right)$$

The $\frac{d}{dx}(\dots)$ notation is quite handy, because it behaves as a function and allows us to write the original expression and the derivative within a single equation:

$$\frac{d}{dx}\left(x^3 + 1\right) = 3x^2$$

Therefore, if we wanted to differentiate y twice, we'd do:

$$\frac{d}{dx}\left(\frac{d}{dx}(y)\right)$$
$$= \frac{d^2}{dx^2}(y)$$
$$= \frac{d^2y}{dx^2}$$

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Examples

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If $y = 3x^5 + \frac{4}{x^2}$, find $\frac{d^2y}{dx^2}$.

If $f(x) = 3\sqrt{x} + \frac{1}{2\sqrt{x}}$, find $f''(x)$.

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Test Your Understanding

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If $y = 5x^3 - \frac{x}{3\sqrt{x}}$, find $\frac{d^2y}{dx^2}$.

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3 Given that $y = (2x - 3)^3$, find the value of x when $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$.

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4\quad $\mathrm{f}(x) = px^3 - 3px^2 + x^2 - 4$. When $x = 2$,
$\mathrm{f}''(x) = -1$. Find the value of p.